



BRACT IT Change Request Document

Title	Controlling Ceiling Increased Loan disbursement — Progoti Program API Changes		
CR ID	BBMFv2-5852		
JIRA ID	BBMFv2-5852		
Created by	Md Farhan Razy	Business Systems Analyst	
Reviewed by	Syed Sami	Business Systems Analyst	
Approved by			



Revision History

Date	Version	Created by	Description of change
28.12.2025	1.0.0	MD Farhan Razy	Initial draft
06.1.2026	1.0.1	MD Farhan Razy	Updated draft
27.1.2026	1.0.2	MD Farhan Razy	Updated based on the review session
9.2.2026	1.0.3	MD Farhan Razy	Added two new api, and loan pre validation api



Purpose

The purpose of this change request is to implement configurable ceiling-increase rules and approval workflows for repeat loans under the Progoti project. This enhancement will ensure that loan amount increases follow policy guidelines, trigger the correct recommendation and approval hierarchy, and provide transparent percentage calculations within the system. The change removes hardcoded logic, improves operational accuracy, and supports quick policy updates through configurable ceiling limits.

Users

- DCS Users

Scope of this CR

This section provides a clear outline of what falls within the scope of the proposed changes and what is excluded from consideration.

Scope Identification	Description
In Scope	The scope includes: • Introducing repeat_loan_percentage, repeat_ceiling_condition, effective_date and repeat_loan_approver field in Approver-wise Loan Limit Mapping API for DCS validation. • Implementing post-validation rules based on repeat loan ceiling percentage. • Validating approver and recommender layers, checking repeat loan amount, and displaying the highest eligible limit.
Out of Scope	• Any new API development beyond the required enhancements. • Branch-level configuration (ERP does not support branch-specific loan setup). • Projects other than Progoti and ADP. • Changes to approve hierarchy structures. • New loan products or approval roles • Any workflow changes outside of approval/recommendation validation. • Piloting activities (as ceiling change is a global configuration).

Workflow

Not required for this CR.



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List of Features & Stories

Feature ID	Feature Title	Story ID	JIRA ID	Story Title
1	Approval wise loan limit Mapping			
2	Buffer Loan Proposal Validation API (Post validation)			

Features

- Approval wise loan limit Mapping | < Feature ID> | <Feature JIRA ID>

1.1 Approval wire loan limit Mapping API Changes | < Story ID> | <Story JIRA ID>

- User Story:
As an authorized user, I want the Approval wire loan limit Mapping API to return *the repeat_loan_percentage, effective_date and repeat_loan_approver*. so that the client system (DCS) can correctly get the loan product configuration information from the system.
- User Journey:
 - ERP Admin configures Increase Condition for Repeat Loans (%) in:
 - ERP → Microfinance → Programme Admin → Loan Policy → Role Wise Approver Limit
 - DCS calls the Approval-wise Loan Limit Mapping API.
 - API returns all above-limit fields along with the new configured *repeat_loan_percentage, effective_date and repeat_loan_approver*.
 - DCS Pre Validation & Post Validation use this field in ceiling calculations and role validation.
- Actors: ERP, QSoft/DCS, MF Program Admin
- Scope:
 - Addition of a new field in the API response: *repeat_loan_percentage, effective_date and repeat_loan_approver*. Provide this value for all approver limit mappings under Progoti (and ADP).
- Pre-condition:
 - ERP must configure "Increase Condition for Repeat Loans (%)", Special condition repeat loan ceiling is checked, effective_date and repeat loan approver is defined for the project.
 - For Progoti & ADP only.
- Impacted Areas:
 - Approval-wise Loan Limit Mapping API
 - Buffer Loan Proposal (Post Validation) API

- API

API Details

URL	Version	Description
{BASE_URL}/smart-mf/dcs/v*/approver-wise-loan-limit-mappings	<i>New Version will be provided</i>	Returns role-wise approval limits and the new <i>repeat_loan_percentage, effective_date, repeat_loan_approver</i> parameter used for ceiling validation.

- Request Body

Method	GET
Request	This request does not have a body
Response	<pre>[{ "approverId": 3, "approverName": "BM", "projectCode": "60", "branchGrowthType": "HIGH", "limitFrom": 5000, "limitTo": 75000, "repeatLimitFrom": 5000, "repeatLimitTo": 120000, "repeat_loan_percentage": 40, "effective_date": "2024-05-12", "repeat_loan_approver": 2, "groupCategory": null }]</pre>



Request Body

This request does not have a body

Requirements

New Field Requirements

Field Additions (The following new fields will be introduced)

- repeat_loan_percentage: Integer
- effective_date: String
- repeat_loan_approver: Integer

Business Logic

- When repeat_ceiling_condition = True:
 - The system will populate predefined (preset) values for:
 - repeat_loan_percentage
 - effective_date
 - repeat_loan_approver
- When repeat_ceiling_condition = False:
 - The default value for the above fields (repeat_loan_percentage, effective_date, repeat_loan_approver) will be null
- If ERP sets value = 0:
 - Ceiling logic disabled
 - DCS treats repeat loan ceiling as N/A
 - Approval of hierarchy default rule applies

Backward Compatibility

- Existing fields remain unchanged.
- App must not break for consumers expecting old response.
- New field is additive
- A new API version will be provided for this API.

Acceptance Criteria

01. New Field Returned

API response must contain repeat_loan_percentage, effective_date, repeat_loan_approver for Progoti and ADP projects.

02. Ceiling Logic Enabled/Disabled

- If value > 0 → DCS applies ceiling rules
- If value is not configured or zero = null (API will return) → DCS disables repeat loan ceiling validation

03. DCS Uses This Value Correctly

- Pre Validation uses the field for ceiling computation
- Post Validation enforces recommender/approver role

04. Backward Compatible

- No failures in existing DCS screens/services

05. Data Accuracy

- API must return the exact value configured in ERP UI

1.2 All Branch Wise Loan Products API Changes

User Story:

As an authorized system user, I want the All Branch-Wise Loan Products API to return the repeat_ceiling_condition for each loan product per branch. So that (DCS) can correctly determine whether repeat-loan ceiling rules are applicable for a given product.

User Journey:

- QSoft/DCS sends a request to API.

Actors: QSoft/DCS

Scope:

- All Branch Wise Loan Products API (New Fields only)

Pre-condition:

- Repeat loan validation (Does follow special repeat loan approval condition) must be configured for the loan product in ERP.
- Loan products must be available and mapped to branches in the system.

Impacted Areas:

- Mapping API response only.

API

API Details

URL	Version	Description
-----	---------	-------------



{BASE_URL}/smart-mf/dcs/v2/all-branch-wise-loan-products	V2 (New)	- Requested changes will be incorporated in Version 2 - Version 1 will remain as-is
--	----------	--

Request Body

No Changes required

Response Body

Existing Response Format	Proposed Response format
[{ "productId": 31, "productCode": "BD-10027", "productName": "BD-10027 General 18% Reducing Prottasha Loan(Monthly)", "projectCode": "279", "memberCategoryName": "ERMG Member", "memberCategoryId": 31, "approverLimitSetupType": 1 },..... {...}]	[{ "productId": 31, "productCode": "BD-10027", "productName": "BD-10027 General 18% Reducing Prottasha Loan(Monthly)", "projectCode": "279", "memberCategoryName": "ERMG Member", "memberCategoryId": 31, "repeat_ceiling_condition ": "True", "approverLimitSetupType": 1, }, { "productId": 0000, "productCode": " BD-10001", "productName": " New_Shadhin Loan 24% (Single)", "projectCode": "000", "memberCategoryName": "xxxx", "memberCategoryId": 00, "repeat_ceiling_condition ": "False", "approverLimitSetupType": 2, }, {...}]

Request Parameter Fields:

- The request will be as-is.

Response Parameter Fields:

Property Name	Value Type	Description
repeat_ceiling_condition	Boolean	Is repeat loan condition Applicable for Loan [Default value = false] Box checked=" true" Box Unchecked=" false"

- The response will have a new field "repeat_ceiling_condition = Boolean", Which can have boolean values [true/ false]
 - ⌚ true = Box Checked (repeat ceiling applicable)
 - ⌚ false = Box Unchecked (repeat ceiling not applicable)

Mock-up/ Prototypes:

Not required.

Requirements:

- New Field: repeat_ceiling_condition
 - o Type: Boolean
- Accepted Values:
 - o true → Repeat ceiling is applicable
 - o false → Repeat ceiling is not applicable
- Null or empty values are NOT accepted.
- The field must be present for every product/plan across all branches.

Acceptance Criteria:

- API returns repeat_ceiling_condition for all products/plans per branch.
- The field is mandatory, and null/empty/invalid values are not allowed.

1.3 Specific Branch Wise Loan Products API Changes



User Story: As an authorized user, I want the Specific Branch-Wise Loan Products API to return the *repeat_ceiling_condition* so that the client system (DCS) can correctly determine whether repeat-loan ceiling rules are applicable for a given product.

User Journey:

- QSoft/DCS sends a request to API.

Actors: QSoft/DCS

Scope:

- All Branch Wise Loan Products API (New Fields only)

Pre-condition:

- Repeat loan validation (Does follow special repeat loan approval condition) must be configured for the loan product in ERP.
- Loan products must be available and mapped to branches in the system.

Impacted Areas:

- 1.1 Mapping API response only.

API

- API Details

URL	Version	Description
{BASE_URL}/smart-mf/dcs/v2/specific-branch-wise-loan-products	V2 (New)	- Requested changes will be incorporated in Version 2 - Version 1 will remain as-is

Request Body

No Changes required

Response Body

Existing Response Format	Proposed Response format
[{ "productId": 78, "productCode": "BD-10074", "productName": "General 24 % Reducing (Medical Loan)", "projectCode": "225", "memberCategoryName": "Associate Member", "memberCategoryId": 11, "branchCode": "0227", "approverLimitSetupType": 1 },..... {....}]	[{ "productId": 31, "productCode": "BD-10027", "productName": "BD-10027 General 18% Reducing Prottasha Loan(Monthly)", "projectCode": "279", "memberCategoryName": "ERMG Member", "memberCategoryId": 31, "branchCode": "0227", "repeat_ceiling_condition": "False", "approverLimitSetupType": 1 }, { "productId": 0000, "productCode": " BD-10001", "productName": " New_Shadhin Loan 24% (Single)", "projectCode": "000", "memberCategoryName": "xxxx", "memberCategoryId": 00, "branchCode": "0000", "repeat_ceiling_condition": "True", "approverLimitSetupType": 2 }, {....}]

Request Parameter Fields:

- The request will be as-is.

Response Parameter Fields:

Property Name	Value Type	Description
repeat_ceiling_condition	Boolean	Is repeat loan condition Applicable for Loan [Default value = false] Box checked=" true" Box Unchecked=" false"

- The response will have a new field "repeat_ceiling_condition = Boolean", Which can have boolean values [true/ false]
 - ⌚ true = Box Checked (repeat ceiling applicable)
 - ⌚ false = Box Unchecked (repeat ceiling not applicable)



Requirements:

- New Field: repeat_ceiling_condition
 - Type: Boolean
- Accepted Values:
 - true → Repeat ceiling is applicable
 - false → Repeat ceiling is not applicable
- Null or empty values are NOT accepted.
- The field must be present for every product/plan across all branches.

Acceptance Criteria:

- API returns repeat_ceiling_condition for all products/plans per branch.
- The field is mandatory, and null/empty/invalid values are not allowed.

Buffer Loan Proposal API [Post Validation] | < Feature ID> | <Feature JIRA ID>

Buffer Loan Proposal API Changes | < Story ID> | <Story JIRA ID>

- User Story:

As an authorized user, I want the system to validate repeat loan ceiling percentage and the approver hierarchy during post validation so that loan proposals exceeding the configured ceiling follow the correct approval workflow and are prevented from being saved in the buffer if unauthorized.
- User Journey:
 - QSoft/DCS sends a POST request to the Buffer Loan Proposal API.
 - The system validates the request fields against a predefined set of rules. If the request satisfies all validation criteria, it is placed into the buffer.
 - The system checks:
 - Ceiling increase percentage compared to previous disbursed loan(s).
 - Whether the proposed repeat loan amount is within the configured ceiling limit.
 - Whether the requester falls under the correct approver hierarchy based on increase percentage.
 - If all validations pass, the request proceeds.
 - If any validation fails, an appropriate error is returned.
- Key changes in business:

Ceiling Increase Limit

 - For repeat loans, the new repeat loan amount cannot increase by more than or equal 40% compared to the previous loan amount.

Approval Rules for >=40% Increase

If the increase is more than or equal 40%, special approval steps apply:

 - a. For AAM/AM authority loans:
 - Approval will be given by RM/SRM.
 - b. For RM/SRM authority loans:
 - Approval will be given by the PM/SPM.

Hierarchy will be based on the following setup.

Hierarchy: PO < ABM < BM < AAM < AM < ARM < RM < SRM < DM < SM < PH < HO < Director

For this scope, only follow the approver logic up to PM/DM, if the approver of last disbursed loan is PM, then if repeat loan exceeds 40% of previous disbursed amount and the amount is within PM limit, then the approver will be PM, will not be escalated to SM

The conditions will be checked in following steps:

- 1st step: Repeat Loan Amount check (if repeat loan proposal => 40% ceiling increase of previous loan or not)
- 2nd step: Repeat Loan Proposal => 40% ceiling increase of previous loan is true, then the system will check the approver of the latest disbursed loan
 - ✓ The system will check the Repeat Loan Approver Limit Range and based on that set the Approver Limit.
 - If previous loan approver is AAM/AM and the proposed amount is within Repeat Loan Approver Limit of RM, then the approver will be set to RM.
 - If previous loan approver is AAM/AM and the proposed amount exceeds Repeat Loan Approver Limit of RM, then the approver will be set to DM if the proposed amount is within DM approver limit.
 - If previous loan approver is RM and the proposed amount is within Repeat Loan Approver Limit of DM, then the approver will be set to DM.
 - If previous loan approver is AAM/AM and the proposed amount exceeds Repeat Loan Approver Limit of DM, then the approver will be set to SM/PH/HO/Director matching with the Repeat Loan Approver Limit configuration.
 - ✓ Repeat Loan Proposal => 40% ceiling increase of previous loan is false, then the system will follow the approver from Role Wise Approver Limit.



- If the proposed loan amount ceiling increase is less than 40%, then the system will allow the existing Approver in Role Wise Approver Limit to approve the Repeat Loan.

Actors: QSoft/DCS

Scope:

- Buffer Loan Proposal API (Repeat Loan Ceiling Validation + Approver Layer Validation Only).

Pre-condition:

- Ceiling increase limits must be available in Approver-wise Loan Limit Mapping configuration.
- If no ceiling limit is set, the system should follow the regular approval rule set in ERP.
- Previous loan disbursement history must be available in the system to calculate ceiling %.
- The proposed Repeat Loan Product Has Repeat Loan Validation activated. Post validations will only apply when the proposed Loan product is **enabled for Repeat Loan Validation {repeat_ceiling_condition ": True".}**
- Member has a previous active or closed loan.
- Repeat loan proposal is more than 40% ceiling increased than previous latest loan.

Loan History Basis

- Only the **enabled for Repeat Loan Validation** for Progoti General Loan will be used for repeat loan ceiling calculation.
- Top up loans are excluded from this calculation.
- If the member has no previous loan, ceiling validation is not applicable.

Ceiling Layer Applicability

Ceiling validation and approval layer checks apply only for AAM/AM & RM/SRM role users

Higher authorities (e.g., PM/SPM and above) are exempt from this ceiling validation layer.

Hierarchy will be based on the following setup. PO < ABM < BM < AAM < AM < ARM < RM < SRM < DM < SM < PH < HO < Director

- $CeilingIncrease\% = (proposedLoanAmount - lastClosedGeneralLoanAmount) / lastClosedGeneralLoanAmount * 100$
- **Impacted Areas:**
 - Buffer Loan Proposal API request.

• **API**

API Details

URL	Version	Description
{BASE_URL}/smart-mf/dcs/v*/buffer-loan-proposals	<i>To be decided</i>	The Loan Proposal API must validate repeat loan ceiling percentage and verify recommender/approver eligibility.

Request Body

Method	POST
Request	<pre>[{ "id": "b6b4545f-486e-4eee-8a14-609e25112410", "void": null, "orgId": 2, "voCode": "", "updated": false, "memberId": 55272756, "nominees": [{ <nominee_object> }, { <nominee_object> }], "wantsFireInsurance": true, "fireInsuranceId": 2, "fireInsuranceDetails": { <fireInsurance_object> }, "schemeld": 330, "sectorId": 43, "updatedAt": "2024-12-04T09:30:00+05:30", "branchCode": "5662", "consentUrl": null, "frequencyId": 3,</pre>



	<pre>"projectCode": "60", "signConsent": false, "subSectorId": 24, "memberTypeId": 1, "policyTypeId": 1, "coBorrowerDto": null, "loanAccountId": 0, "loanProductId": 76, "premiumAmount": 300, "secondInsurer": null, "loanProposalId": 44161393108225, "microInsurance": true, "applicationDate": "2025-06-16", "modeOfPaymentId": null, "rejectionReason": "", "disbursementDate": null, "approvedLoanAmount": 200000, "insuranceProductId": 37, "loanApprover": 4, "proposedLoanAmount": 200000, "proposedGrantAmount": 0, "loanProposalStatusId": 2, "approvedDurationInMonths": 9, "proposalDurationInMonths": 9 }</pre>
Response	<pre>{ "message": "OK", "code": "200" } { "type": "about:blank", "status": 400, "instance": "/dcs/v1/buffer-loan-proposal-validations", "timestamp": "2024-09-11T11:47:03.009342124", "httpStatus": 400, "errorMessage": "Bad Request", "errors": [{ "message": "Approval limit exceeded. The maximum amount you can approve is {{(MAX_APPROVE)}}. This loan requires approval from {{(NEXT_APPROVER_ROLE)}}.", },], }, { "type": "about:blank", "status": 400, "instance": "/dcs/v1/buffer-loan-proposal-validations", "timestamp": "2024-09-11T11:47:03.009342124", "httpStatus": 400, "errorMessage": "Bad Request", "errors": [{ "message": "When ((loanApprover": null,)) we will show a validation message, that approver value can't be Null.", },], },], },</pre>

Request Parameter Fields:

- The request will be as-is.

Response Parameter Fields:

- The response will be as-is. 2 new validations will be added where we will check the loan wise approver value, When the **proposed loan amount** is greater than repeat loan ceiling value and when a loan proposal request is received with "loanApprover": null (or missing).



- Mock-up/ Prototypes:
Not required.
- Requirements:
 - The Repat Loan Approver can be selected based on the approver hierarchy. Always the Repeat Loan Approver will be the next level in the hierarchy when the proposal exceeds the percentage of Increased Condition for Repeat Loans.
 - Current system Hierarchy will be based on the following setup. **PO < ABM < BM < AAM < AM < ARM < RM < SRM < DM < SM < PH < HO < Director**
(For this scope, only consider AAM/AM > RMSRM and RM/SRM > PM/SPM. All other levels of approver will not be applicable)
 - When a new loan is proposed, system should check if the member has any previous or current loan. If a member has multiple closed or current loans under the Product where Does Follow Special Repeat Loan Approver Conditions is checked in Loan Product Setup. (For this scope, only Progoti/ADP General Loan Products), **then Repeat loan percentage will be calculated based on the latest general loan (closed/active) amount.**
 - Then the system will check the ceiling increase of the previous loans.
 - If the proposed loan amount ceiling increase is **equal or greater than 40%** of previous loan amount, then the Approver will be next in hierarchy from the hierarchy setup

The conditions will be checked in following steps:

- **1st step:** Repeat Loan Amount check (if repeat loan proposal => 40% ceiling increase of previous loan or not)
- **2nd step:** Repeat Loan Proposal => 40% ceiling increase of previous loan is true, then the system will check the approver of the latest disbursed loan
 - ✓ The system will check the Repeat Loan Approver Limit Range and based on that set the Approver Limit.
 - If previous loan approver is **AAM/AM** and the proposed amount is within Repeat Loan Approver Limit of **RM**, then the approver will be set to **RM**.
 - If previous loan approver is **AAM/AM** and the proposed amount exceeds Repeat Loan Approver Limit of **RM**, then the approver will be set to **DM** if the proposed amount is within **DM** approver limit.
 - If previous loan approver is **RM** and the proposed amount is within Repeat Loan Approver Limit of **DM**, then the approver will be set to **DM**.
 - If previous loan approver is **AAM/AM** and the proposed amount exceeds Repeat Loan Approver Limit of **DM**, then the approver will be set to **SM/PH/HO/Director** matching with the Repeat Loan Approver Limit configuration.
 - ✓ Repeat Loan Proposal => 40% ceiling increase of previous loan is false, then the system will follow the approver from Role Wise Approver Limit.
- If the proposed loan amount ceiling increase is less than 40%, then the system will allow **the existing Approver in Role Wise Approver Limit to approve the Repeat Loan.**

Ceiling Configuration

- System must read repeat loan ceiling % from:
ERP → Programme Admin → Approver-wise Loan Limit Mapping (repeat_loan_percentage)
- Ceiling must be fully configurable by MF Tech.

Ceiling Calculation

- $\text{CeilingIncrease\%} = ((\text{proposedLoanAmount} - \text{lastLoanAmount}) / \text{lastLoanAmount}) * 100$
- If $\text{CeilingIncrease\%} \geq \text{configured limit}$ (according to ERP business) → system must also compare against previous loan
- System must determine $\text{comparisonBasis} = \text{lastLoan}$
- In case of error, the loan proposal must not be saved or proceed to the next workflow stage.
- Validation must work for only repeat loans.

Approval Layer Rules

- If increase < ceiling → Same as regular ERP business (normal hierarchy applies)
 - If increase >= ceiling:
 - AAM/AM** level loan will be approved by RM/SRM (for some of the branches if SRM is not available, for those loan approver will be RM vice versa)
 - RM/SRM** level loan will be approved by DM
 - *(For this scope, only consider AAM/AM > RMSRM and RM/SRM > PM/SPM. All other levels of approver will not be applicable)*
 - **"loanApprover":** null value will not be acceptable value need to be integer (1,2, 3..),
 - This change should not impact/change/remove any existing rules.
 - Approver wise loan limit business for lien loan will remain as-is.
- **Validation Rule (Post Validation):**
 - **Scenario 1:**
 - **Title:** Post-Validation for Repeat Loan Approval (Ceiling & Role Limits)
 - **Context:** When a repeat loan proposal is submitted for approval, the system must enforce two constraints before proceeding:
 - **Repeat Loan Ceiling** — The requested amount must not exceed the **dynamic ceiling** (previous loan × (1 + ceiling%)).



- **Role-wise Approver Limit** — The approver's **role-based limit** must not be exceeded. If either validation fails, the **Post Validation API** should block the action and return a standardized error response with a **short validation text** showing the **maximum approvable amount** and **Next approver authority (who can approve this loan now)** based on the failing rule (ERP Business configuration).
- **Approver layer validation Condition:** When the **proposed loan amount** is greater or equal than repeat loan ceiling value system will show the validation Error Message. If the Role wise approver limit validation fails then following error message should be shown in response with proper error code, system will show a validation text

- **Error Message:**

```
{
  "type": "about:blank",
  "status": 400,
  "instance": "/smart-mf/dcs/v3/buffer-loan-proposals",
  "timestamp": "2026-01-06T05:48:02.000Z",
  "httpStatus": 400,
  "errorMessage": " Bad Request.",
  "errors": [
    {
      "message": "Approval limit exceeded. The maximum amount you can approve is {{MAX_APPROVE}}. This loan requires approval from {{NEXT_APPROVER_ROLE}}. ",(Keep same message in OTC as well)
    }
  ]
}
```

- **Scenario 2:**

- **Context:** During Post Validation (server-side enforcement before approval), a request is received with "loanApprover": null (or missing).
- **DCS Need:** Approval action must have an identified approver. Null or missing approver is invalid.
- **Outcome:** Block the operation and return a clear, standardized error response.
- **Field:** loanApprover
- **Rule:** Must be non-null, any integer value
- **Failure Condition:** loanApprover == null

Error message:

- When ("loanApprover": null,) we will show a validation message, that approver value cant be Null,

```
{
  "type": "about:blank",
  "status": 400,
  "instance": "/smart-mf/dcs/v3/buffer-loan-proposals",
  "timestamp": "2026-01-06T05:48:02.000Z",
  "httpStatus": 400,
  "errorMessage": " Bad Request.",
  "errors": [
    {
      "message": " Loan Approver cannot be null."
    }
  ]
}
```



}

Response body value:

- Where: $\text{MAX_APPROVE}} = \text{previousLoanAmount} \times (1 + \text{ceilingPercent})$
- $\text{MAX_APPROVE}}$: Double value
- $\text{NEXT_APPROVER_ROLE}}$ will be Special conditional repeat loan approver (according to ERP business)
- $\text{NEXT_APPROVER_ROLE}}$: String value (AAM, AM, RM, PM)
- CeilingPercent is dynamic, fetched from ERP configuration.

For Example:

- Previous Loan = 100,000
- Ceiling = 30% (dynamic)
- Repeat Loan Ceiling = $100,000 \times 1.30 = 130,000$
- Requested = 150,000
- For this scenario the error message will be: "Approval limit exceeded. The maximum amount you can approved is 130,000 This loan requires approval from RM ",

- **Acceptance Criteria:**

01. Ceiling % Validation

System must calculate ceiling % and determine whether the proposed loan is within the configured limit.

02. Approver Layer Validation

System must ensure that users act only within their allowed role (Recommend vs Approve).

03. Reject If Unauthorized

If ceiling limit is exceeded and user role is not authorized to approve, API rejects request.

04. Comparison with Last Loan

If ceiling exceeded, system must compute and show percentage compared with previous loan.

- System checks if member has a previous active or closed loan under Progoti/ADP general products.
- System checks if the proposed loan product is Progoti/ADP general product.
- System calculates ceiling increase from previous loans.
- If ceiling increase is $\geq 40\%$, system uses hierarchy-based Loan Approver from Role Wise Approver Limit.
- If ceiling increases $< 40\%$, system uses existing approver for Repeat Loans.

Buffer Loan Proposal Validation API [Pre Validation] | < Feature ID> | <Feature JIRA ID>

Buffer Loan Proposal Validation API Enhancement | < Story ID> | <Story JIRA ID>

- **User Story:** As an authorized user, I want the system to validate the repeat loan ceiling percentage and the approver layer before proposal submission, so that loan amounts exceeding the configured ceiling follow the correct approval workflow and unauthorized actions are blocked early.
- **User Journey:**
 - QSoft/DCS sends a POST request to the Buffer Loan Proposal validation API.
 - The system validates the request fields against a predefined set of rules. If the request satisfies all validation criteria, it is placed into the buffer.
 - **The system checks:**
 - Ceiling increase percentage compared to previous disbursed loan(s).
 - Whether the proposed repeat loan amount is within the configured ceiling limit.
 - Whether the requester falls under correct approver hierarchy based on increase percentage.
 - If all validations pass, the request proceeds.
 - If any validation fails, an appropriate error is returned.
- Key changes in business:

Ceiling Increase Limit

 - For repeat loans, the new repeat loan amount cannot increase by more than or equal 40% compared to the previous loan amount.

Approval Rules for $\geq 40\%$ Increase

If the increase is more than or equal 40%, special approval steps apply:

- a. For AAM/AM authority loans:
 - Approval will be given by RM/SRM (based on the availability of the role in a area).



- b. For RM/SRM authority loans:
 - Approval will be given by the DM.

Hierarchy will be based on the following setup.

Hierarchy: PO < ABM < BM < AAM < AM < ARM < RM < SRM < DM < SM < PH < HO < Director

For this scope, only follow the approver logic up to **PM/DM**, if the approver of last disbursed loan is **PM**, then if repeat loan exceeds 40% of previous disbursed amount and the amount is within PM limit, then the approver will be PM, will not be escalated to SM

- If the proposed loan amount ceiling increase is less than 40%, then the system will allow the existing Approver in Role Wise Approver Limit to approve the Repeat Loan.
- **Actors:** QSoft/DCS
- **Scope:**
 - Pre Validation (Repeat Loan Ceiling & Approval Layer Validation only).
- **Pre-condition:**
 - Configurable ceiling limit (repeat_loan_percentage) is present in Approver-wise Loan Limit Mapping
 - Previous disbursed loan data is available.
 - Applies to AAM/AM and RM/SRM authority loans.
 - ERP keeps existing validations; this API adds ceiling and hierarchy checks
- **Impacted Areas:**
 - Buffer Loan Proposal API request.
 - Pre Validation service
 - Approval-wise Loan Limit Mapping API (new field repeat_loan_percentage)

• **API**

API Details

URL	Version	Description
{BASE_URL} /smart-mf/dcs/v*/buffer-loan-proposal-validations	<i>To be decided</i>	The buffer loan proposal validation should validate the “Role wise Approver Limit”

Request Body

Method	POST
Request	<pre>[{ "id": "b6b4545f-486e-4eee-8a14-609e25112410", "void": null, "orgId": 2, "voCode": "", "updated": false, "memberId": 55272756, "nominees": [{ <nominee_object> }, { <nominee_object> }], "wantsFireInsurance": true, "fireInsuranceId": 2, "fireInsuranceDetails": { <fireInsurance_object> }, "schemeId": 330, "sectorId": 43, "updatedAt": "2024-12-04T09:30:00+05:30", "branchCode": "5662", "consentUrl": null, "frequencyId": 3, "projectCode": "60", "signConsent": false, "subSectorId": 24, "memberTypeId": 1, "policyTypeId": 1, "coBorrowerDto": null, "loanAccountId": 0,</pre>



	<pre>"loanProductId": 76, "premiumAmount": 300, "secondInsurer": null, "loanProposalId": 44161393108225, "microInsurance": true, "applicationDate": "2025-06-16", "modeOfPaymentId": null, "rejectionReason": "", "disbursementDate": null, "approvedLoanAmount": 200000, "insuranceProductId": 37, "loanApprover": Null, "proposedLoanAmount": 200000, "proposedGrantAmount": 0, "loanProposalStatusId": 2, "approvedDurationInMonths": 9, "proposalDurationInMonths": 9 }</pre>
Response	<pre>{ "message": "OK", "code": "200" } { "type": "about:blank", "status": 400, "instance": "/dcs/v1/buffer-loan-proposal-validations", "timestamp": "2024-09-11T11:47:03.009342124", "httpStatus": 400, "errorMessage": "Bad Request", "errors": [{ "message": "Approval limit exceeded. The maximum amount you can approve is {{MAX_APPROVE}}. This loan requires approval from {{NEXT_APPROVER_ROLE}}.", "code": "L-308" }] },</pre>

Request Parameter Fields:

- The request will be as-is.

Response Parameter Fields:

- The request will be as-is. 2 new validations will be added with the existing validation.

Error Table:

SL no	Case	Validation Message	Validation Code
1	When the proposed loan amount is greater than repeat loan ceiling value system will show the validation Error Message.	Approval limit exceeded. The maximum amount you can approve is {{MAX_APPROVE}}. This loan requires approval from {{NEXT_APPROVER_ROLE}}.	L-308

Requirements:

- The Repat Loan Approver can be selected based on the approver hierarchy. Always the Repeat Loan Approver will be the next level in the hierarchy when the proposal exceeds the percentage of Increased Condition for Repeat Loans.
- Current system Hierarchy will be based on the following setup. **PO < ABM < BM < AAM < AM < ARM < RM < SRM < DM < SM < PH < HO < Director**
(For this scope, only consider AAM/AM > RMSRM and RM/SRM > PM/SPM. All other levels of approver will not be applicable)
- When a new loan is proposed, system should check if the member has any previous or current loan. If a member has multiple closed or current loans under the Product where Does Follow Special Repeat Loan Approver Conditions is checked in Loan Product Setup. (For this scope, only Progoti/ADP General Loan Products), **then Repeat loan percentage will be calculated based on the latest general loan (closed/active) amount.**
- Then the system will check the ceiling increase of the previous loans.



- If the proposed loan amount ceiling increase is **equal or greater than 40%** of previous loan amount, then the Approver will be next in hierarchy from the hierarchy setup

The conditions will be checked in following steps:

- **1st step:** Repeat Loan Amount check (if repeat loan proposal => 40% ceiling increase of previous loan or not)
- **2nd step:** Repeat Loan Proposal => 40% ceiling increase of previous loan is true, then the system will check the approver of the latest disbursed loan
 - ✓ The system will check the Repeat Loan Approver Limit Range and based on that set the Approver Limit.
 - If previous loan approver is **AAM/AM** and the proposed amount is within Repeat Loan Approver Limit of **RM**, then the approver will be set to **RM**.
 - If previous loan approver is **AAM/AM** and the proposed amount exceeds Repeat Loan Approver Limit of **RM**, then the approver will be set to **DM** if the proposed amount is within **DM** approver limit.
 - If previous loan approver is **RM** and the proposed amount is within Repeat Loan Approver Limit of **DM**, then the approver will be set to **DM**.
 - If previous loan approver is **AAM/AM** and the proposed amount exceeds Repeat Loan Approver Limit of **DM**, then the approver will be set to **SM/PH/HO/Director** matching with the Repeat Loan Approver Limit configuration.
 - ✓ Repeat Loan Proposal => 40% ceiling increase of previous loan is false, then the system will follow the approver from Role Wise Approver Limit.
- If the proposed loan amount ceiling increase is less than 40%, then the system will allow **the existing Approver in Role Wise Approver Limit to approve the Repeat Loan.**

Ceiling Configuration

- System must read repeat loan ceiling % from:
ERP → Programme Admin → Approver-wise Loan Limit Mapping (repeat_loan_percentage)
- Ceiling must be fully configurable by MF Tech.

Ceiling Calculation

- $\text{CeilingIncrease\%} = ((\text{proposedLoanAmount} - \text{lastLoanAmount}) / \text{lastLoanAmount}) * 100$
- If CeilingIncrease% >= configured limit (according to ERP business) → system must also compare against previous loan
- System must determine comparisonBasis = lastLoan
- In case of error, the loan proposal must not be saved or proceed to the next workflow stage.
- Validation must work for only repeat loans.

Approval Layer Rules

- If increase < ceiling → Same as regular ERP business (normal hierarchy applies)
- If increase >= ceiling:
 - AAM/AM** level loan will be approved by RM/SRM (for some of the branches if SRM is not available, for those loan approver will be RM vice versa)
 - RM/SRM** level loan will be approved by DM
- *(For this scope, only consider AAM/AM > RMSRM and RM/SRM > PM/SPM. All other levels of approver will not be applicable)*
- "loanApprover": null value will be applicable for pre validation.
- **Handling loanApprover Value in Pre Validation API** During Pre Validation, the loanApprover field will be treated as *optional*.

This means:

- If (loanApprover = null) → No validation will be triggered, and the system will NOT block the request.
- If (loanApprover contains any integer value) → The system must validate the approver limit exactly the same way as in Post Validation.
- This change should not impact/change/remove any existing rules.
- Approver wise loan limit business for lien loan will remain as-is.

- **Validation Rule (Pre Validation):**

- **Scenario 1:**
 - **Title:** Post-Validation for Repeat Loan Approval (Ceiling & Role Limits)
 - **Context:** When a repeat loan proposal is submitted for approval, the system must enforce two constraints before proceeding:
 - **Repeat Loan Ceiling** — The requested amount must not exceed the **dynamic ceiling** (previous loan × (1 + ceiling%)).
 - **Role-wise Approver Limit** — The approver's **role-based limit** must not be exceeded. If either validation fails, the **Post Validation API** should block the action and return a standardized error response with a **short validation text** showing the **maximum approvable amount** and **Next approver authority (who can approve this loan now)** based on the failing rule (ERP Business configuration).
 - **Approver layer validation Condition:** When the **proposed loan amount** is greater or equal than repeat loan ceiling value system will show the validation Error Message. If the Role wise approver limit validation fails then following error message should be shown in response with proper error code, system will show a validation text
 - **Error Message:**

```
{
  "type": "about:blank",
  "status": 400,
  "instance": "/smart-mf/dcs/v3/buffer-loan-proposals",
  "timestamp": "2026-01-06T05:48:02.000Z",
  "httpStatus": 400,
  "errorMessage": " Bad Request.",
  "errors": [
    {
      "message": "Approval limit exceeded. The maximum amount you can approve is {{MAX_APPROVE}}. This loan requires approval from {{NEXT_APPROVER_ROLE}}. ",
    }
  ]
}
```

Scenario 2: Handling loanApprover Value in Pre Validation API

During Pre Validation, the loanApprover field will be treated as *optional*.

This means:

If (loanApprover = null) → No validation will be triggered, and the system will NOT block the request.

If (loanApprover contains any integer value) → The system must validate the approver limit exactly the same way as in Post Validation.

Rule 1 — loanApprover is null → Skip Validation

- Condition:
loanApprover == null
- Action:
 - Skip all approver-related validations
 - Return success (continue other validations normally)
- Rationale:
In Pre Validation, the system may not yet have an approver; it's not mandatory at that stage.

Rule 2 — loanApprover contains an integer → Apply Regular Validation

- Condition:
loanApprover != null AND value is integer
- Action:
Perform full approver validation, identical to Post Validation:
 - Check if the approver has the required approval role
 - Check role-wise approval limit
 - If requested amount exceeds approver's limit → show error

Response body value:

- Where: $\text{MAX_APPROVE}} = \text{previousLoanAmount} \times (1 + \text{ceilingPercent})$
- $\text{MAX_APPROVE}}$: Double value
- $\text{NEXT_APPROVER_ROLE}}$ will be Special conditional repeat loan approver (according to ERP business)
- $\text{NEXT_APPROVER_ROLE}}$: String value (AAM, AM, RM, PM)
- CeilingPercent is dynamic, fetched from ERP configuration.

For Example:

- Previous Loan = 100,000
- Ceiling = 30% (dynamic)
- Repeat Loan Ceiling = $100,000 \times 1.30 = 130,000$
- Requested = 150,000
- For this scenario the error message will be: "Approval limit exceeded. The maximum amount you can approved is 130,000 This loan requires approval from RM ",



- **Note: error code might change while development**
- Where: $\{\{MAX_APPROVE\}\} = \text{previousLoanAmount} \times (1 + \text{ceilingPercent})$
- **ceilingPercent** is dynamic, fetched from configuration.

For Example:

- Previous Loan = 100,000
- Ceiling = 30% (dynamic)
- Repeat Loan Ceiling = $100,000 \times 1.30 = 130,000$
- Requested = 150,000

- **Mock-up/ Prototypes:**
Not required.
- **Acceptance Criteria:**
 - LoanApprover value null will be acceptable in pre validation
 - System checks if member has a previous active or closed loan under Progoti/ADP general products.
 - System checks if the proposed loan product is Progoti/ADP general product.
 - System calculates ceiling increase from previous loans.
 - If ceiling increase is $\geq 40\%$, system uses hierarchy-based Loan Approver from Role Wise Approver Limit.
 - If ceiling increases $< 40\%$, system uses existing approver for Repeat Loans.

• Risk Analysis

This section evaluates potential project risks by assigning a Risk Level (High/Medium/Low), assessing the Likelihood of the Event, and outlining a Mitigation Plan. It aims to proactively identify and manage uncertainties to ensure effective risk mitigation throughout the project lifecycle.

SL	Risk	Level (High/Medium/Low)	Likelihood of Event	Mitigation Plan
1.	Missing or null repeat_loan_percentage in API response may break ceiling calculation	Medium	Medium	

• Change Effort Estimation

1. Effort Estimation

The Effort Estimation section provides a detailed breakdown of the anticipated effort required for each activity given the successful implementation of the proposed changes. This includes estimating the amount of time in hours and the total billable hour will be considered as the final invoice.

SL	Item	Days
1	Project Management	
2	Business Analysis	
3	Development	
4	SQA	
5	Implementation	
	Total	

Note: If any

CR Accumulation

CR Initiated by	[Insert Client Name]	
CR Initiated on		
Type of CR	Category of CR	Accumulation (%)
Regular ☐	New Addition ☐ Full Replacement ☐ Partial Modification ☐	X%
Ad-Hoc (No accumulation) ☐		
Data Entry (No accumulation) ☐		
Data Correction (No accumulation) ☐		



Note: Effort given in the "Regular" type of CR and Category in "New Addition" and "Partial Modification" will be included in AMC as per Accumulation percentage.